

NeckErgonomics: Adapting Head Rotation in HMDs with Predicted Neck Muscle Activity

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In HMDs, every head movement is work for the neck

1

The head is the controller

We **point, scan** windows and look around 360° content by turning the head.

2

The headset adds load

Extra mass in front of the face raises the work of the neck muscles.

3

Sessions are getting longer

XR is moving from short demos to work, training and entertainment.

Reducing neck strain isn't just about comfort but also about usability and accessibility.

Adaptive interfaces help, but infer effort indirectly

A promising way to reduce this load is **adaptive XR interaction**: interface parameters such as cursor gain or display position change at runtime to reduce effort.

1 Reposition content

Move content toward the forward view, so the head can return to a neutral pose.

McGill et al. 2020; O'Hagan et al. 2025

2 Modulate head-motion gain

Amplify head rotation, so less physical turning is needed to reach the same view.

Jay & Hubbard 2003; Langbehn et al. 2019; Lee et al. 2024

However, these adaptations infer physical demand from

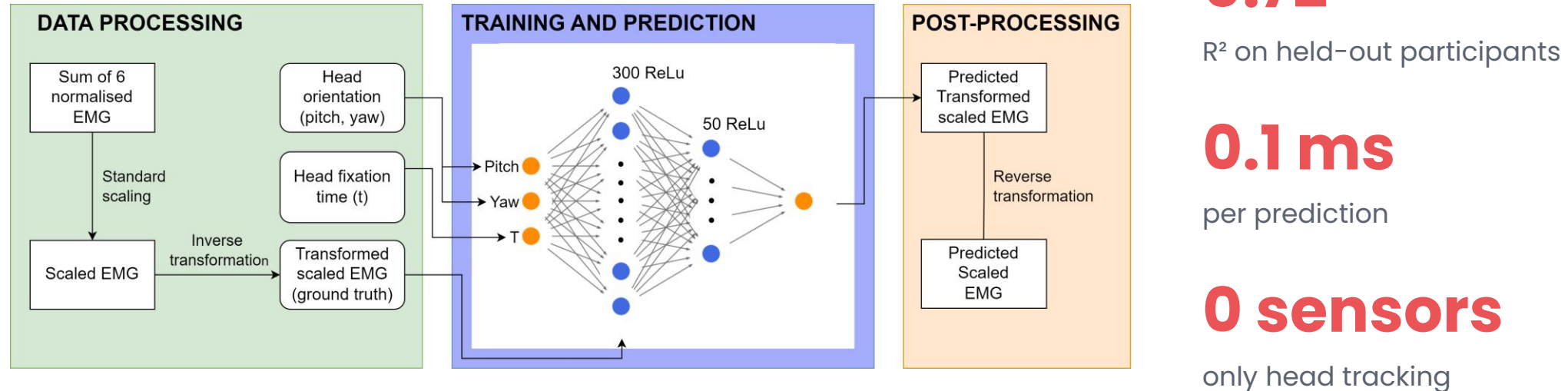
Geometry

Movement

Task heuristics

rather than **from neck muscle activity**

We trained a model on our neck EMG dataset



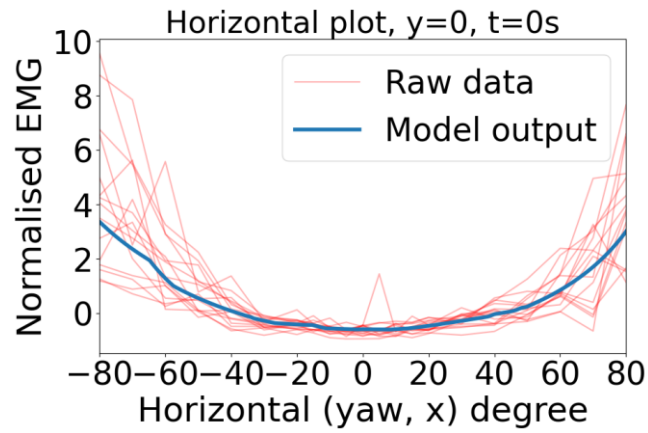
Data Our open neck EMG dataset [1]: 15 people, 6 neck channels, $\pm 80^\circ$ yaw, $\pm 50^\circ$ pitch, with fixation time.

Model A small neural network (MLP): head yaw, pitch and fixation time in, predicted neck EMG out.

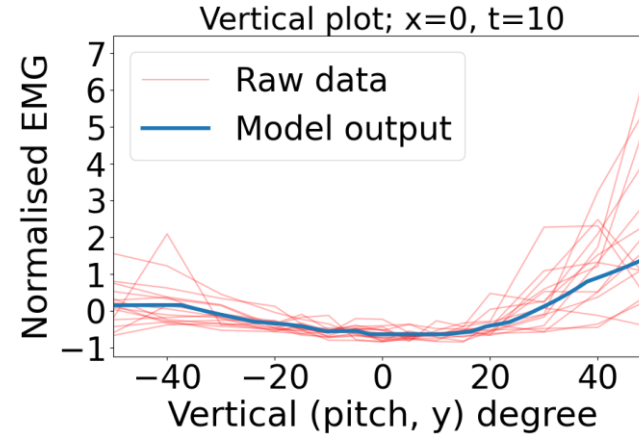
Result No sensors needed at runtime: the headset's head tracking is enough.

[1] G. Li, F. Weidner, J. Hu, H. Gellersen. 2025. Quantifying neck muscle activity during head fixation in VR. *Frontiers in Virtual Reality* 6.

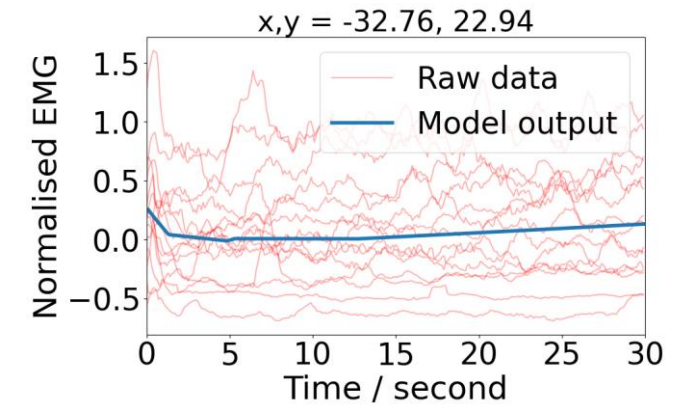
Predicted EMG: bowl-shaped, settling over time



Yaw sweep: low near centre, rising at the sides



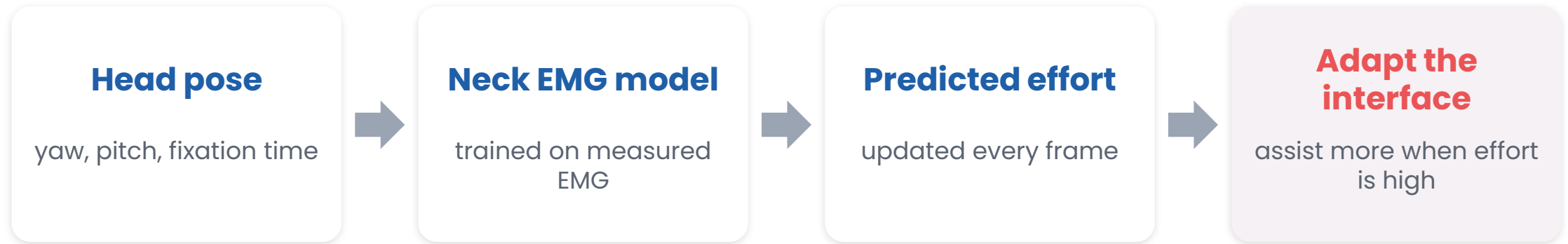
Pitch sweep: same bowl shape



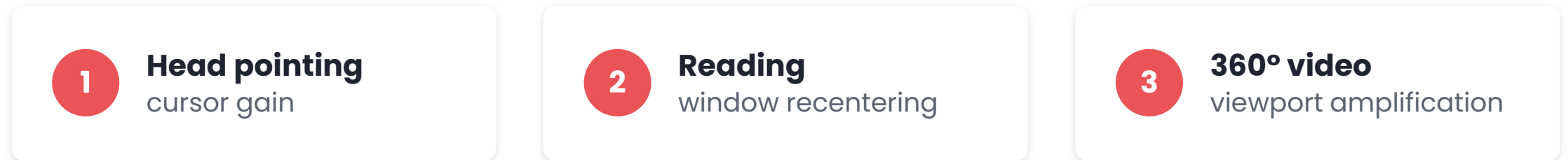
Fixed pose: drops over ~3 s, then stable

Blue: model output · red: participant data

NeckErgonomics: adapt to predicted neck effort

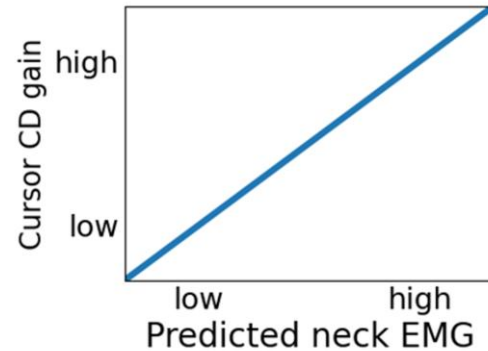
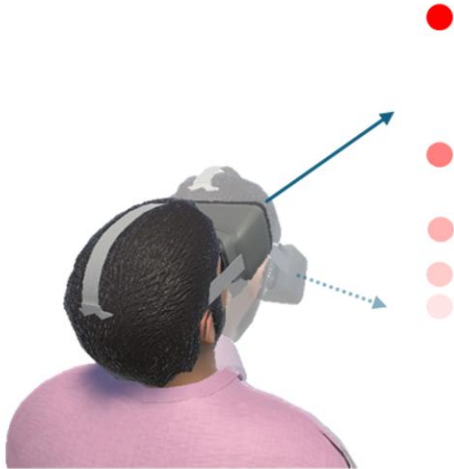


One signal, three interaction techniques

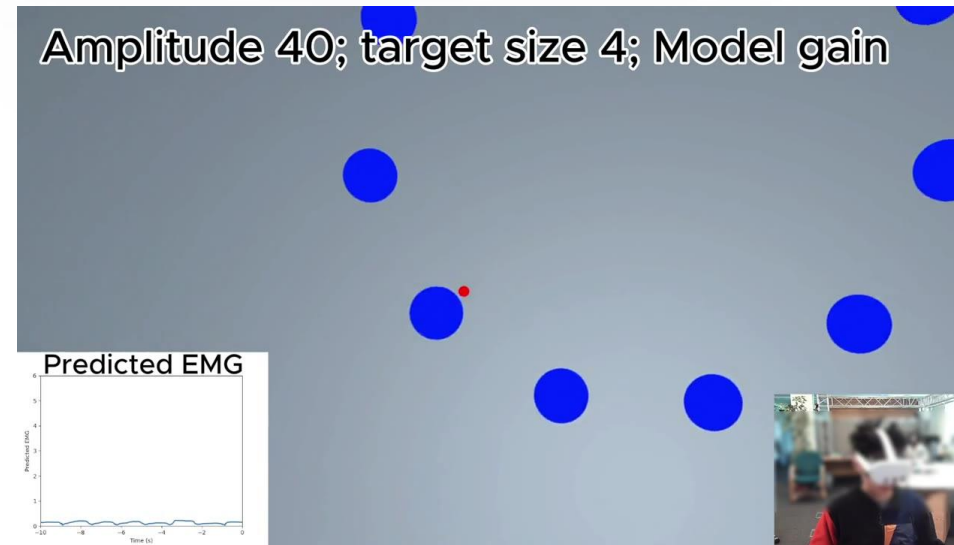


Higher predicted effort → **stronger assistance** · near the comfort zone → **normal behaviour**

Head pointing: the cursor helps when turning is costly



Predicted effort → cursor gain



Study recording, 22 s

Near the centre

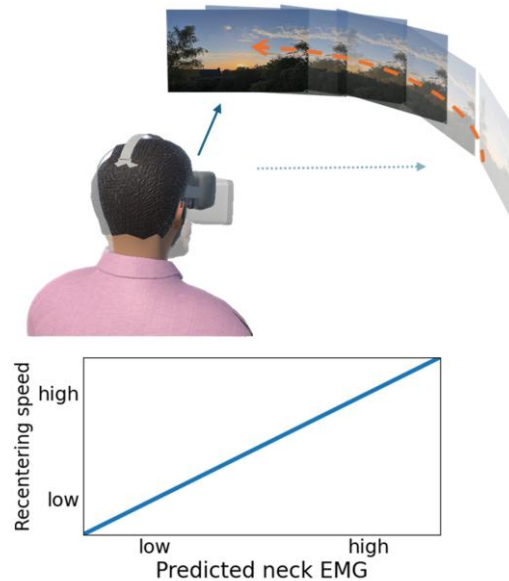
The cursor follows the head 1:1, so pointing stays precise.

Turning far out

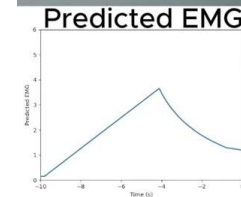
The cursor moves further than the head, so targets are reached with less rotation.

Reading: windows come to you when the pose is costly

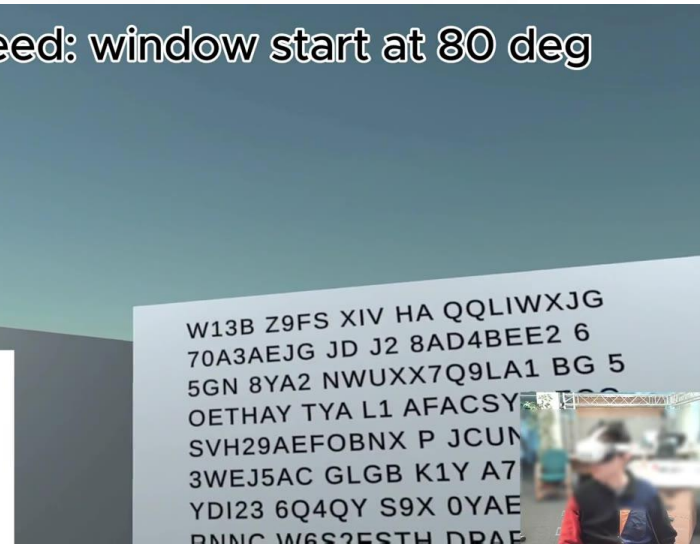
Predicted effort + time → recentering speed



Model speed: window start at 80 deg



Study recording, 22 s



A quick glance

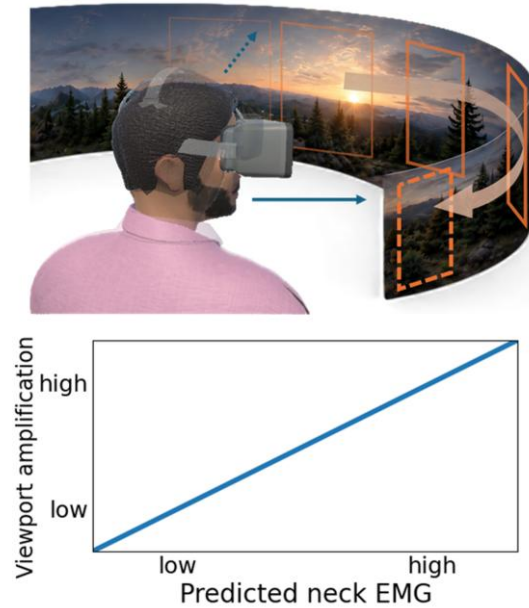
The window stays where it is; short looks to the side are fine.

Reading off to the side

The window glides toward the front, faster when the pose is demanding, slower near the centre.

360° video: the view turns further than the head

Predicted effort → viewport amplification



Study recording, 22 s

Looking ahead

The view moves 1:1 with the head, which keeps viewing comfortable.

Turning to the side

The view rotates more than the head, so you can look behind you without turning all the way.

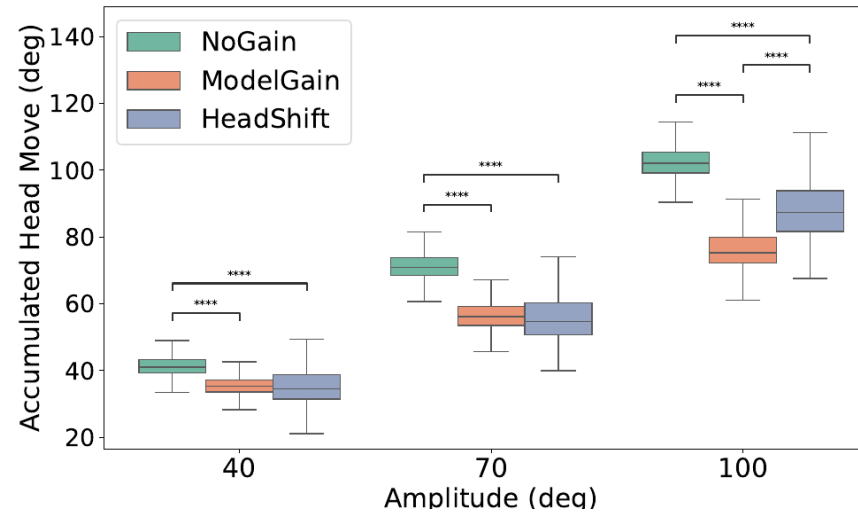
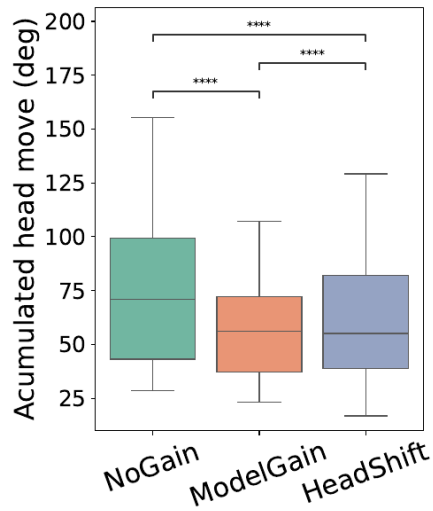
Study: one within-subjects session, three tasks

N = 18 · seated · Meta Quest 3 · ~90 min

Task	Ours	Baselines	Key measures
Head pointing (Fitts-law like)	ModelGain	NoGain (1:1), HeadShift	Head movement, time, errors, throughput
Reading in a side window ($\pm 40^\circ$, $\pm 80^\circ$)	ModelSpeed	NoSpeed, ConstantSpeed ($3^\circ/\text{s}$)	Head rotation, time, errors, comfort
360° video (4 min)	ModelGain	NoGain (1:1), HeadGain (1:2)	NASA-TLX, VRSQ, head angles

We did not record EMG during the study: head movement, sustained rotation and perceived demand are proxies.

Pointing: less head movement, but slower



Less head movement

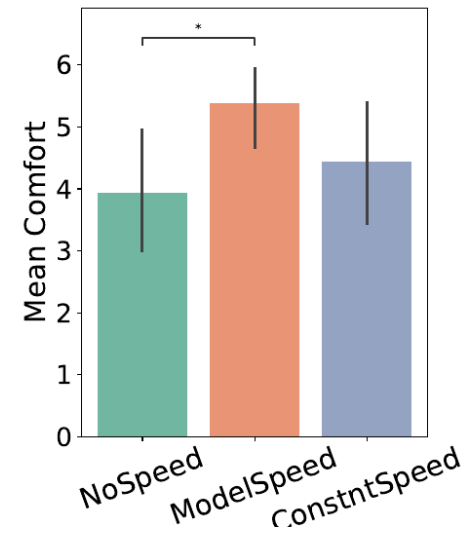
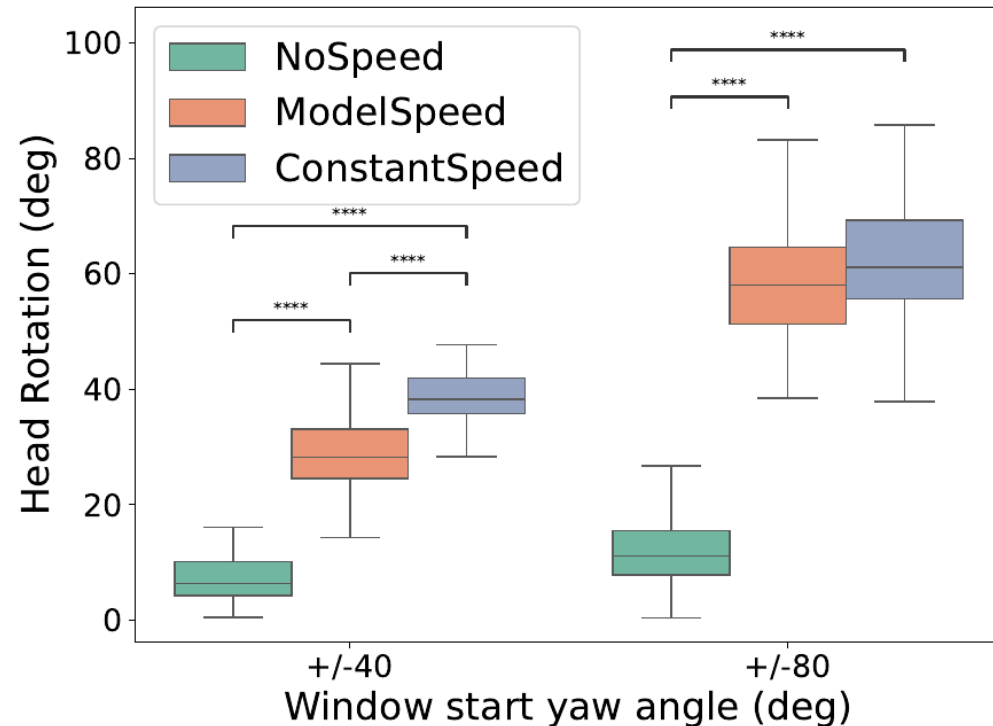
ModelGain < HeadShift < NoGain; the gap is widest at 100° amplitude.

Trade-off

Slower completion, more errors on 2° targets than HeadShift, lower throughput than NoGain.

Accumulated head movement by technique (left) and target amplitude (right)

Reading: less head rotation, more comfort



Both adaptive techniques cut head rotation.

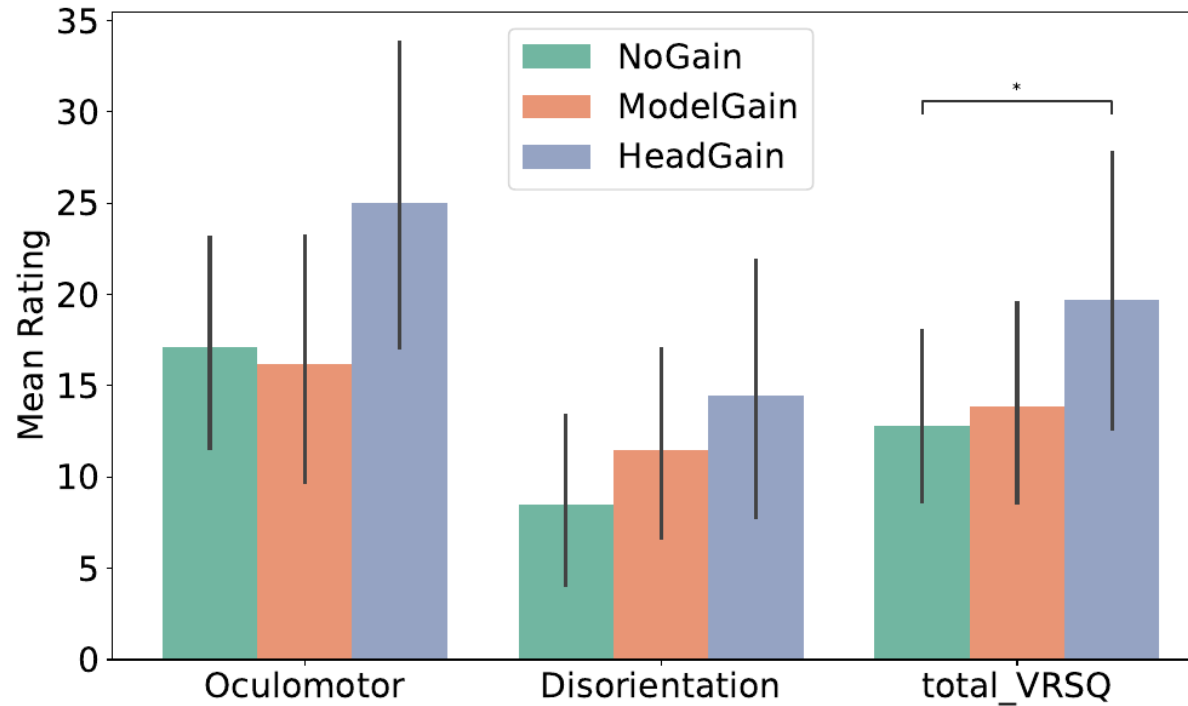
At $\pm 40^\circ$, ModelSpeed beat ConstantSpeed.

Comfort: **5.4 vs 3.9** for ModelSpeed vs NoSpeed (7-pt).

ConstantSpeed was faster; errors did not differ.

Head rotation by window start angle (left) · mean comfort with 95% CI (right)

360° video: less demand, no extra sickness



VRSQ scores, 95% CI

- 1 **Lower physical demand**
ModelGain vs NoGain on NASA-TLX.
- 2 **No extra sickness**
HeadGain raised VRSQ above NoGain; ModelGain did not.
- 3 **Full 360° coverage**
Head angles stayed within about $\pm 100^\circ$, vs $\pm 170^\circ$ with NoGain.

What we learned

1

Ergonomics by design

Neck cost becomes part of the interaction logic, not something we measure afterwards.

2

One signal, many techniques

The same predicted EMG drove cursor gain, recentering speed and viewport rotation.

3

Effort is a trade-off

Each task paid differently: speed, precision, reading stability or sickness. The scale k sets how hard the system intervenes.

Limitations and next steps

Limitations

One model for everyone: no per-user strength or mobility

No live EMG in the study; trained on HTC Vive Pro Eye data, used on Quest 3

Young, healthy participants; seated, short lab sessions, one stimulus per task

Next

Validate predictions against live EMG

Personalise the model to individual users

Longer sessions: fatigue and discomfort over time

Older adults and people with limited mobility



Thank you!

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